

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 3.4: Specialized Architectures & Curve-Fitting Case

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Topic 3.4 Agenda

1. **Specialized Architectures** — ICNN, monotone, theory-consistent
2. **Case Study** — curve fitting with PyTorch
3. **Lab Preview** — Lec 3 lab walkthrough (Lec03_Lab_ML_Basics)

Arc: *T1 Why ML → T2 NN Concepts → T3 Training + Backprop → T4 Specialized + Lab*

Specialized Architectures

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Why Standard FFNNs Fall Short in Quant Macro

- **Nested “Russian Doll” structure:** Quantitative macro models involve *multiple* nested optimization problems.
- **Typical hierarchy:**
 1. **Outer Loop:** Market clearing / calibration (solve for prices p^*)
 2. **Middle Loop:** Value function iteration (solve Bellman equations)
 3. **Inner Loop:** Agent optimization (choose c, ℓ, k' given state and prices)
- **Problem:** Approximating value functions or costs with generic FFNNs turns the *inner* problem into a non-convex optimization
 - Multiple local optima and unstable outer iterations
 - Economic misspecification (e.g., upward-sloping demand, non-concave value functions)

Idea: Inject Economic Structure into the Network

Key idea: Design architectures that *respect* economic shape restrictions *by construction*.

Property	Mathematical Gain	Economic Payoff
Convexity / Concavity	Unique global optimum	Stable policy rules / VFI
Monotonicity	Invertible mappings	Well-behaved supply/demand
Homogeneity	$f(\lambda x) = \lambda f(x)$	Consistency with balanced growth

- This turns the neural net from a **black box** into a **glass box**: flexible enough to fit data, but disciplined by theory.

Convolution: Sliding a Small Filter Over the Input

Three objects (first conv layer, classifying a handwritten digit):

- **Input** X — the image: a grid of numbers (e.g. a 28×28 intensity map); in deeper layers, the previous layer's feature map.
- **Filter / kernel** W — a *small* grid of **learned** weights (e.g. 3×3): a “pattern template.”
- **Output** Y — the **feature map**: one number per location, scoring how strongly W 's pattern appears there.

What convolution does: slide W across X ; at each spot overlay the filter, multiply overlapping cells, and add them (a local weighted dot product):

$$\underbrace{Y(i, j)}_{\text{output}} = \sum_m \sum_n \underbrace{X(i - m, j - n)}_{\text{input patch}} \underbrace{W(m, n)}_{\text{filter weight}}$$

$Y(i, j)$ is **large** where the local patch of X *looks like* the pattern in W (an edge, a curve, a stroke).

Weight sharing: the same tiny W is reused everywhere — it finds its feature *wherever* it occurs (translation equivariance), with far fewer parameters than a dense layer.

Pooling — and Why a CNN Is a “Learned Manifold Map”

Pooling: shrink and summarize. Replace each small window R of the feature map with one summary value; **max-pooling** keeps the strongest response:

$$\text{MaxPool}(i, j) = \max_{(m, n) \in R} X(m, n)$$

- **Downsamples** (less compute) and adds **translation invariance** — “it’s a 7” wherever in the frame the 7 sits.

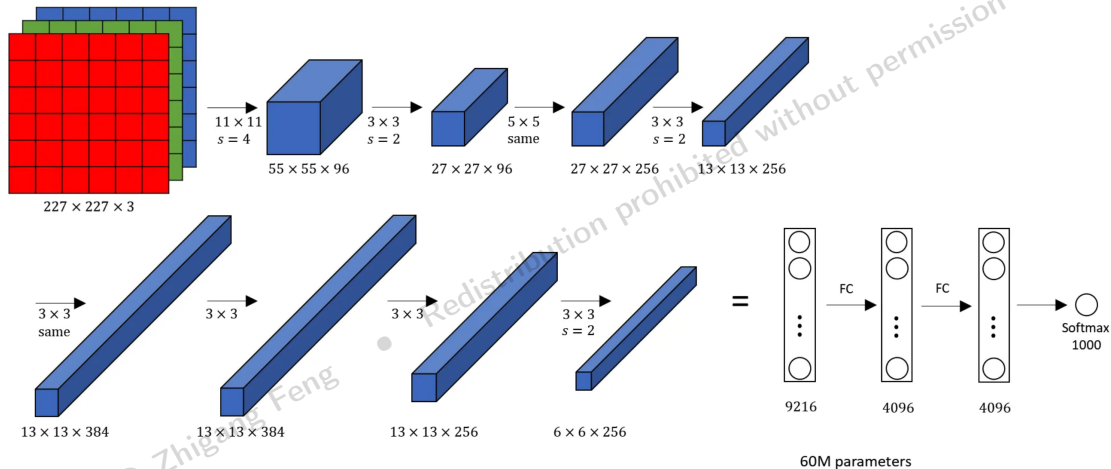
Stacking [convolution $\rightarrow$ nonlinearity $\rightarrow$ pooling] builds a hierarchy: *pixels* $\rightarrow$ *edges* $\rightarrow$ *motifs* $\rightarrow$ *objects*.

The two-stage view: the stack is a *learned* map π compressing the raw, high-dimensional input onto a compact, low-dimensional feature vector (a *manifold*); a head g_ϕ reads off the prediction:

$$\mathbf{x} \xrightarrow{\pi \text{ (conv+pool)}} \mathbf{z} \xrightarrow{g_\phi \text{ (head)}} \hat{\mathbf{p}}$$

Economic analogy: π learns a *sufficient statistic* — like reducing a high-dimensional macro state to the few moments that drive decisions (mean wealth in Krusell–Smith).

AlexNet: A Landmark CNN Architecture



Krizhevsky, Sutskever & Hinton (2012, NeurIPS) — 15.3% top-5 error vs. 26.2% for non-DL. Launched the DL agenda

Lec. 4 turns onto dynamic macro.

Input Convex Neural Networks (ICNNs)

- **Motivation:** In economics, many functions are known to be convex or concave: cost functions, duality mappings, value functions (after sign change).
- **ICNN architecture** (Amos, Xu & Kolter 2017, ICML): ensure $f(x; \theta)$ is convex in x by construction:

$$\mathbf{z}^{(0)} = x$$
$$\mathbf{z}^{(\ell+1)} = \sigma \left(\underbrace{W^{(\ell)}}_{\geq 0} \mathbf{z}^{(\ell)} + A^{(\ell)} x + b^{(\ell)} \right), \quad \ell = 1, \dots, L-1$$

- **Key constraints:**
 - Weights $W^{(\ell)} \geq 0$ (element-wise non-negative) for $\ell \geq 1$
 - Activation σ must be convex and non-decreasing (ReLU, softplus)
 - "Pass-through" connections $A^{(\ell)} x$ can have unconstrained weights
- **Result:** The composition of convex, non-decreasing functions with non-negative weights preserves convexity.
- **For concavity:** Use $g(x) = -f(x; \theta)$ where f is an ICNN. *Lec. 4 T4 revisits with*

ICNN: Proof Sketch (Why It's Convex)

We prove by induction that each $z_l(x)$ is convex in x .

Step 1: Base Case

- $z_0(x) = x$ is linear; any linear function is both convex and concave.

Step 2: Inductive Step

$$z_{l+1}(x) = \sigma(W_z^{(l)} z_l(x) + W_x^{(l)} x + b_l)$$

- By induction, $z_l(x)$ is convex.
- With $W_z^{(l)} \geq 0$: $W_z^{(l)} z_l(x)$ is a non-negative sum of convex functions $\Rightarrow$ convex.
- Adding linear term $W_x^{(l)} x + b_l$ preserves convexity.
- Applying convex, non-decreasing $\sigma(\cdot)$ to a convex argument preserves convexity.

Therefore: All layers $z_l(x)$, and hence $f(x)$, are convex in x . (Lec. 4 T4 uses this directly on the Bellman value function.)

Do Constraints Limit Expressivity?

The concern: By restricting $W_z \geq 0$, do we lose the universal approximation property?

Theorem (Chen, Shi, & Zhang, 2019):

An Input Convex Neural Network can approximate any convex function over a compact domain to arbitrary accuracy.

Reference: Chen, Y., Shi, Y., & Zhang, B. (2019). *Optimal Control via Neural Networks: A Convex Approach*. International Conference on Learning Representations (ICLR).

Intuition:

- A maximum of affine functions ($\max_i \{a_i^\top x + b_i\}$) is convex and can approximate any convex shape.
- An ICNN with ReLU activations acts as a hierarchical max-affine approximator.

Economist's take: You impose a **shape restriction** (convexity) without imposing a specific parametric form (CES, Cobb-Douglas). Architecture encodes theory; data still choose the function inside the theory-consistent class. (*Lec. 4 T4 deploys this in dynamic equilibrium.*)

Partial ICNN (PICNN): The Macro Reality

The issue: Value functions $V(k, z)$ are typically:

- **Concave** in endogenous states k (capital, assets)
- **Non-concave / arbitrary** in exogenous states z (TFP, income shocks)

A standard ICNN would incorrectly enforce convexity on z as well.

Solution: Partial ICNN (PICNN, Amos et al. 2017) — two paths combined so that convexity holds in k for every fixed z :

1. **Free path for z :** $u_l = \tanh(\widetilde{W}_u^{(l)} u_{l-1} + \widetilde{b}_u^{(l)})$ (unconstrained FFNN of z).
2. **Convex path for k , gated by z :**

$$z_{l+1} = \sigma \left(\underbrace{W_z^{(l)} (z_l \odot u_l)}_{W_z^{(l)} \geq 0} + W_x^{(l)} k + b^{(l)} \right)$$

where $\odot$ is element-wise multiplication; u_l acts as a z -dependent positive gate on z_l .

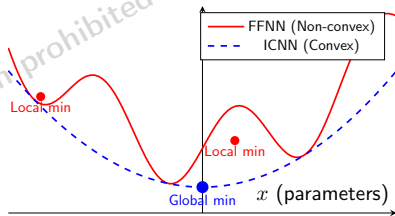
Convexity in k : at fixed z , u_l is constant. Non-negative scaling, adding linear-in- k , and applying convex non-decreasing σ all preserve convexity.

Landscape Geometry: Why FFNNs Get Stuck

Standard FFNN loss surface:

- Rugged, non-convex.
- Many local minima.
- Outcome depends on initialization.

Implication: Re-train the same model with a different seed, get a different fit.
Hard to audit; hard to publish.



Why Convexity Matters for Macro Solvers

- **Bellman / Euler equations have a unique fixed point** (under standard contraction conditions). A solver that converges to a *local* minimum of the residual is not converging to that fixed point.
- **Reproducibility:** For a published paper, the policy function should not depend on the random seed. Convex training removes that dependence.
- **HA models compound the problem:** Aiyagari and Krusell-Smith require thousands of inner solves at every outer iteration. A 5% chance of a bad local min per inner solve becomes a near-certain failure mode at scale.
- **Validation cost:** Without convexity guarantees, every solution needs a hand-checked sanity audit (analytical limit, Euler residual heatmap — Lec. 4 T3). With convexity, validation is once at the architecture level.

The ICNN Advantage: Bounding Optimization Risk

- **Restated formally:** An ICNN makes $f(x; \theta)$ convex in its *input* x by construction, so *inference* (optimizing/inverting over x at fixed θ) is a convex problem with a unique global optimum. Training over the parameters θ remains non-convex (cf. T3's non-convex loss landscape); convexity is a property of the learned function, not of the loss surface. For the convex inference subproblem, gradient descent's worst case is $O(1/T)$ convergence (Nesterov 2018, *Lectures on Convex Optimization*).
- **Reference theorem:** Chen, Shi & Zhang (2019), ICLR, "Optimal Control via Neural Networks: A Convex Approach" — shows that the ICNN-parameterized optimal control problem has the same convex structure as the underlying linear-quadratic regulator.
- **Macro corollary:** You can swap a hand-coded convex solver (Sirignano & Spiliopoulos 2018 (DGM), Maliar et al. 2021 deep-learning Euler method) for an ICNN-based solver without losing convergence guarantees. The architecture provides the math; PyTorch provides the speed.
- **Cost:** Restricted expressivity (Lec. 3 T4 "Do Constraints Limit Expressivity?"). Partial ICNN (Amos et al. 2017) recovers most of it.

ICNN: Robust PyTorch Implementation

Strategy: Use reparameterization (Softplus) instead of clamping to enforce $W_z \geq 0$.

```
1 class ICNN(nn.Module):
2     def __init__(self, d_in, hidden_sizes):
3         super().__init__()
4         # Passthrough weights (Input -> Hidden) [unconstrained]
5         self.W_x = nn.ParameterList([
6             nn.Parameter(torch.randn(h, d_in))
7             for h in hidden_sizes])
8         # Recurrent weights (Hidden -> Hidden) [unconstrained params]
9         # Apply Softplus() during forward pass to enforce  $W_z \geq 0$ 
10        self.W_z_params = nn.ParameterList([
11            nn.Parameter(torch.randn(h, h_prev))
12            for h, h_prev in zip(hidden_sizes[1:], hidden_sizes[:-1])])
13        self.biases = nn.ParameterList([
14            nn.Parameter(torch.zeros(h)) for h in hidden_sizes])
15
16    def forward(self, x):
17        z = x
18        for i, (Wz_p, Wx, b) in enumerate(
19            zip(self.W_z_params, self.W_x, self.biases)):
20            Wz = F.softplus(Wz_p) # Enforce non-negativity
21            z_next = F.linear(z, Wz) + F.linear(x, Wx) + b
22            z = F.relu(z_next)
23        return z
```


Monotone Neural Networks

- **Motivation:** Economic theory often implies monotonicity:
 - Demand is non-increasing in price (law of demand)
 - Consumption is non-decreasing in wealth (normal goods)
 - Value functions are non-decreasing in the state
- **Architecture for monotonicity in input x_j :**
 - Constrain all weights connecting from x_j through the network to be **non-negative**
 - Use **non-decreasing** activation functions (ReLU, sigmoid, softplus)
 - Composition of non-decreasing functions is non-decreasing
- **Partial monotonicity:** Can enforce monotonicity w.r.t. *some* inputs while leaving others unconstrained — matching economic theory that demand is monotone in price but non-monotone in income (Giffen goods aside).
- **Practical enforcement:** After each gradient step, project weights to $\mathbb{R}_+$ via $w \leftarrow \max(w, 0)$ or use $w = \exp(\tilde{w})$ reparameterization.

Monotone NN: Architecture — Partial Monotonicity

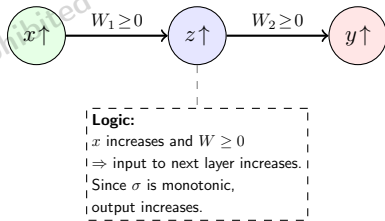
Objective: Ensure $f(x)$ is non-decreasing in selected inputs:

$$x_i \uparrow \Rightarrow f(x) \uparrow$$

Two necessary constraints:

1. **Non-negative weights:** $W^{(l)} \geq 0$ for monotone-input paths (biases unconstrained).
2. **Monotonic activations:** $\sigma'(\cdot) \geq 0$ (ReLU, Sigmoid, Tanh, Softplus).

Partial monotonicity: Some inputs monotone (e.g., income $\rightarrow$ consumption); others free (e.g., interest rate $\rightarrow$ savings). Mirror the partial-ICNN trick.



Monotone NN: Enforcement and Economic Examples

- **Enforcement (same two methods as ICNN):**

- *Clamping*: $W \leftarrow \max(W, 0)$ after each gradient step. Simple but can kill neurons (dead-ReLU analogue).
- *Reparameterization (Softplus)*: train an unconstrained $\widetilde{W}$ and use $W = \text{softplus}(\widetilde{W}) = \log(1 + e^{\widetilde{W}})$. Preferred — gradients flow.

- **Where macro needs monotonicity:**

- *Wage curves* $w(\text{skill})$ — weakly increasing.
- *Consumption functions* $c(\text{wealth})$ — non-decreasing in wealth.
- *Demand curves* $q(p)$ — non-increasing in price (flip sign).
- *Value functions* $V(k)$ in growth models — monotone in capital.

- **PyTorch idiom**: wrap each linear layer's weight as `F.softplus(W_raw)` in the forward call; train normally with AdamW. No special optimizer needed.

Economic Applications of Constrained Architectures

- **ICNN for optimal transport / Wasserstein distance:**

- Kantorovich duality: $W_1(\mu, \nu) = \sup_{f \text{ 1-Lipschitz}} \mathbb{E}_\mu[f] - \mathbb{E}_\nu[f]$
- ICNNs parameterize the convex conjugate — used in generative models and distributional analysis

- **Concave value functions:**

- In many macro models, $V(k)$ is concave in capital
- Approximate V with $-\text{ICNN}(k)$ to enforce concavity *by construction*
- Eliminates spurious non-concavities from unconstrained approximation

- **Monotone policy functions:**

- Savings policy $a'(a, z)$ is typically increasing in current assets a
- Monotone networks guarantee this without post-hoc corrections

- **General principle:** Encoding economic structure into the network architecture reduces the hypothesis space, improves sample efficiency, and produces economically interpretable solutions.

Worked Example: Recovering Preferences via ICNN

Economic problem: Expenditure minimization

$$E(p, u) = \min_{c \geq 0} p \cdot c$$

s.t. $U_\theta(c) \geq u$ (target utility)

The “structured” solution:

- Learn the expenditure function $E_\theta(p, u)$ using a Concave ICNN.
- **Theory:** $E(p, u)$ must be concave in prices p — a structural property of the expenditure function.

Why this is powerful:

1. **Shephard's Lemma + Auto-Diff:** $c^*(p, u) = \nabla_p E_\theta(p, u)$ — demand functions for free.
2. **Integrability:** Because E_θ is structurally concave, the resulting demands satisfy Slutsky-matrix properties by construction.

Same recipe in Lec. 4 T4 with $V(k, z)$ as the macroeconomist's first ICNN.

Choosing the Right Architecture: Cheat-Sheet

Don't use a black box when you have a blueprint.

Architecture	Shape constraint	Where useful in macro	Reuse
FFNN (baseline)	None	Generic regression, forecasting, classification	Lec 3 lab
CNN / AlexNet	Translation invariance	Satellite imagery, spatial data; not core macro	—
ICNN	Input convexity	Bellman equation surfaces, cost & value functions	Lec. 4 T4, Lab 6A/6B
PICNN	Partial convexity in k	Aiyagari, growth model with shocks z	Lec. 4 T4
Monotone NN	Partial monotonicity	Wage curves, consumption rules, demand	Lec. 4 T4
DEQ	Implicit fixed point	Equilibrium-as-layer; HA models	Lec. 6
Dropout (regularizer)	—	Overfitting control	all labs

Takeaway: Structured architectures (shaded) encode economic theory directly — you trade some flexibility for guarantees that match the problem.

Case Study: Curve Fitting and PyTorch

Case Study: DNN vs. Polynomial vs. Chebyshev

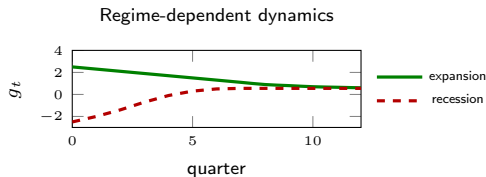
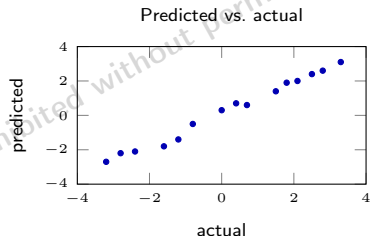
- **Objective:** Approximate a “kinked” function using three methods:
 - Deep Neural Network (DNN)
 - Traditional polynomial approximation (`np.polyfit`)
 - Chebyshev polynomial approximation
- **Why this matters for economists:**
 - Many economic models feature kinks: borrowing constraints, tax brackets, occasionally binding constraints
 - Polynomials suffer from **Gibbs phenomenon** near discontinuities
 - DNNs with ReLU activations handle kinks naturally (piecewise linear)
- **Test function:**

$$y(x) = \begin{cases} \alpha\beta x^\alpha & x \leq 0.5 \\ 2\alpha\beta x^\alpha & x > 0.5 \end{cases}$$

A power function with a discrete jump at $x = 0.5$ — like a policy threshold.

Example: GDP Forecasting with a Tiny MLP

- **Objective:** Predict next-quarter real GDP growth g_t from four lags.
- **Input:** $\mathbf{x}_t = (g_{t-1}, g_{t-2}, g_{t-3}, g_{t-4})$
- **Architecture:** $4 \rightarrow 16$ (ReLU) $\rightarrow 8$ (ReLU) $\rightarrow 1$ (Linear).
- **Why a NN?** GDP dynamics are state-dependent: shocks persist differently in recessions vs. expansions; AR models miss this.
- **Loss:** MSE.



PyTorch: Defining and Training the Model

```
1 import torch
2 import torch.nn as nn
3 import torch.optim as optim
4
5 class GDPNet(nn.Module):
6     def __init__(self):
7         super().__init__()
8         self.net = nn.Sequential(
9             nn.Linear(4, 16), nn.ReLU(),
10             nn.Linear(16, 8), nn.ReLU(),
11             nn.Linear(8, 1)
12         )
13     def forward(self, x):
14         return self.net(x)
15
16 model = GDPNet()
17 criterion = nn.MSELoss()
18 optimizer = optim.Adam(model.parameters(), lr=0.001)
19
20 for epoch in range(1000):
21     optimizer.zero_grad()           # 1. Reset gradients
```

PyTorch: Class-Based Network with Batch Normalization

```
1 class KinkNet(nn.Module):
2     """Network for approximating kinked functions."""
3     def __init__(self, input_dim, hidden_dim1,
4                   hidden_dim2, output_dim=1):
5         super().__init__()
6         self.net = nn.Sequential(
7             nn.Linear(input_dim, hidden_dim1),
8             nn.BatchNorm1d(hidden_dim1),
9             nn.ReLU(),
10            nn.Linear(hidden_dim1, hidden_dim2),
11            nn.BatchNorm1d(hidden_dim2),
12            nn.ReLU(),
13            nn.Linear(hidden_dim2, output_dim)
14        )
15
16    def forward(self, x):
17        return self.net(x)
```

Key additions: BatchNorm1d normalizes activations within each mini-batch — stabilizes training and acts as light regularization.

Lab Preview

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Lab: Machine Learning Basics and Applications

Notebook: Lec03_Lab_ML_Basics.ipynb — one self-contained notebook, three parts.

Part 1: Neural nets as flexible regression

- Train a small PyTorch MLP to learn a consumption function $c = f(w)$ from noisy data — “a neural network is just non-parametric regression.”
- **Universal approximation, hands-on:** build $\hat{f}(x) = a_0 + \sum_i a_i \text{ReLU}(w_i x + b_i)$ and watch ReLU units sum to $\sin x$; then see **implicit regularization** — gradient descent finds the minimum-norm, smooth fit.

Parts 2–3: Two economic applications

- **Predict the 10-year Treasury yield** from the policy rate + unemployment (live FRED data, synthetic fallback), using a proper *time-series* split.
- **Read the Fed's words:** turn hawkish/dovish sentences into bag-of-words features, classify with logistic regression, then a PyTorch MLP.

Preview: What Lec. 4 Does With These Tools

- **Lec. 4 T1 — Motivation + optimal growth setup:** why grids fail in $d \geq 5$; AlphaGo-style intuition for macro state spaces.
- **Lec. 4 T2 — Deep VFI + actor-critic preview:** the NN architectures of T2 + T4 deployed as Bellman solvers; actor-critic algorithm previewed for Lec. 5.
- **Lec. 4 T3 — Euler equation as supervised learning:** the backprop machinery of T3 turned on first-order conditions; PyTorch implementation.
- **Lec. 4 T4 — Structured NNs + validation:** ICNN + Monotone re-applied to dynamic macro with Lab 6A (ground-truth VFI) + Lab 6B (NN solution) + Euler-residual heatmap as the validation gate.

Lec. 5 then formalizes the RL machinery; Lec. 6 scales to heterogeneous-agent models.

Summary

1. **Why ML:** Unstructured data, high-dimensional models, and non-linear dynamics make ML essential for modern macro research.
2. **Neural Networks:** Universal approximators that compose linear transformations with nonlinear activations. Barron (1993): break the curse of dimensionality.
3. **Training:** Forward pass $\rightarrow$ loss $\rightarrow$ backpropagation $\rightarrow$ gradient descent. Mini-batch SGD with Adam is the practical standard.
4. **Regularization:** Weight decay, dropout, early stopping, batch normalization prevent overfitting — the ML analogue of avoiding data mining.
5. **Constrained Architectures:** ICNNs enforce convexity; monotone networks enforce monotonicity. Encoding economic structure into network design improves both accuracy and interpretability.
6. **Next:** Lec. 4 applies these tools to *solve* macro models — deep learning meets the Bellman equation.